Quantum Computing

Quantum Circuits and Teleportation

Lecture 3 Notes

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Prepared for the LebNet and LebSci Quantum Computing Course 2026

About these notes

These notes accompany a nine-lecture course on quantum computing designed and produced by the Lebanese Science Organization. Each chapter corresponds to one lecture. The lectures are concept-focused: they explain what a quantum state is, how entanglement, interference, and measurement combine to produce computation, and what is required to make that computation reliable. Linear algebra, complex numbers, and elementary probability are assumed.

Each chapter names its lecturer and date, opens with a short preface, develops its material with the derivations in place, and closes with a summary. Worked examples appear in numbered boxes, as do definitions, theorems, remarks, and common misconceptions. Works cited in the text are collected in a single reference list at the end of the document.

Conventions used throughout: qubits in a register are labelled from left to right, so that in $|ab\rangle$ the first qubit is a; the first qubit is the top wire in every circuit diagram; the computational basis of two qubits is ordered $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; circuits are read from left to right, so a gate U_1 followed by a gate U_2 acts as the matrix product U_2U_1.

This chapter is exported individually from the full lecture notes (3 of 9). These notes are made available for open, public study. Reuse conditions will be stated separately by the Lebanese Science Organization.

Contents

1 Quantum Circuits and Teleportation **3**
1.1 The circuit model . . . 3
1.1.1 Composition rules . . . 3
1.1.2 Controlled gates . . . 4
1.1.3 Other gates . . . 5
1.1.4 Universality . . . 5
1.2 No-cloning . . . 6
1.3 Teleportation . . . 7
1.3.1 The circuit . . . 7
1.3.2 The teleportation identity . . . 8
1.3.3 Bell-basis measurement . . . 8
1.3.4 Outcomes and corrections . . . 9
1.3.5 No-signalling . . . 10
1.4 Summary . . . 11

References **12**

Chapter 1

Quantum Circuits and Teleportation

Lecturer: Marc Fares

Date: October 10, 2026

Gates, entanglement, measurement, and classical communication each act on quantum information in a different way, and this lecture fixes the notation in which they are combined. It introduces the circuit model and its conventions, the controlled operations that build two-qubit gates from one-qubit ones, and the no-cloning theorem, which forbids copying an unknown state by linearity alone. Teleportation then shows how a shared Bell pair, a Bell-basis measurement, and two classical bits together move an unknown qubit from one place to another, and what each of these ingredients contributes.

1.1 The circuit model

?? introduced the gates X, Z, H acting on one qubit, and **??** introduced the tensor product and the gate CNOT acting on two. A quantum circuit is nothing more than a notation for composing such gates, together with measurements, into a single computation. Its conventions are fixed once and used throughout the course. Each horizontal wire carries one qubit; the first qubit of the register is the top wire. Time runs from left to right, so a box to the left acts before a box to the right. A box labelled U on a wire applies the unitary U to that qubit, and a meter is a measurement in the computational basis whose outcome is a classical bit.

1.1.1 Composition rules

Two rules translate a diagram into a matrix. Gates in sequence on the same wires compose by matrix multiplication in reverse order of appearance: a gate U_1 followed by a gate U_2 is the single unitary U_2U_1, because U_1 acts on the state first and matrices act on the right. Gates on different wires at the same time compose by the tensor product: A on the first qubit together with B on the second is $A \otimes B$, defined on product states by $(A \otimes B)(|a\rangle \otimes |b\rangle) = A\,|a\rangle \otimes B\,|b\rangle$ and extended to all states by linearity. A gate on one wire while the other wire is left alone is $A \otimes \mathbb{I}$ or $\mathbb{I} \otimes B$. Since $(A \otimes \mathbb{I})(\mathbb{I} \otimes B) = A \otimes B = (\mathbb{I} \otimes B)(A \otimes \mathbb{I})$, the relative timing of gates on different wires is immaterial, and only the order along each wire carries meaning.

As an illustration, the Bell preparation circuit of **??**, H on the first qubit followed by CNOT, is the single 4×4 matrix $\mathrm{CNOT}\,(H \otimes \mathbb{I})$. Its four columns are the four Bell states, in the order $|\Phi^+\rangle, |\Psi^+\rangle, |\Phi^-\rangle, |\Psi^-\rangle$ for the inputs $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. The product is written out below, together with the different matrix one obtains by multiplying in the wrong order. Matrices of this

size are rarely multiplied out in these notes; the point of the circuit notation is to reason about the flow of information without doing so.

For matrices of compatible sizes, $(A \otimes B)(C \otimes D) = AC \otimes BD$, and $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$, so a tensor product of unitaries is unitary. In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, the matrix $A \otimes B$ is the block matrix whose (a, a') block is $A_{aa'}B$. Thus

$$H \otimes \mathbb{I} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}, \qquad \mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{1.1}$$

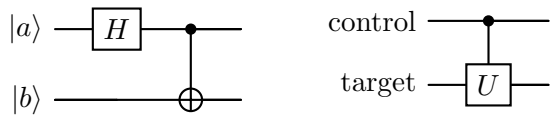
The Bell preparation circuit, H on qubit 1 followed by CNOT, is the product with CNOT on the left. Multiplying by CNOT on the left exchanges the third and fourth rows:

$$\mathrm{CNOT}\,(H \otimes \mathbb{I}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix}. \tag{1.2}$$

The columns are the images of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, namely $|\Phi^+\rangle$, $|\Psi^+\rangle$, $|\Phi^-\rangle$, $|\Psi^-\rangle$, as claimed above. The opposite order, $(H \otimes \mathbb{I})\,\mathrm{CNOT}$, exchanges the third and fourth columns of $H \otimes \mathbb{I}$ instead, and sends $|00\rangle$ to $|+\rangle\,|0\rangle$, a product state. Order matters, and the circuit fixes it: left to right in the diagram is right to left in the product.

1.1.2 Controlled gates

The Bell preparation circuit is drawn on the left below, beside a second two-qubit circuit.



The right-hand diagram is the general controlled operation, which is the one genuinely new construction of this section.

Definition 1.1: Controlled-U gate

Let U be a unitary on one qubit. The controlled-U gate on two qubits, with the first qubit as control and the second as target, is

$$C(U) = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes U = \begin{pmatrix} \mathbb{I} & 0 \\ 0 & U \end{pmatrix}, \tag{1.3}$$

where the block form is written in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. On product states, $C(U)\,|0\rangle\,|\phi\rangle = |0\rangle\,|\phi\rangle$ and $C(U)\,|1\rangle\,|\phi\rangle = |1\rangle\,U\,|\phi\rangle$. In a diagram the control is a dot and the target is the box U, joined by a vertical line.

The gate CNOT is $C(X)$, and its action on basis states, $|a\rangle\,|b\rangle \mapsto |a\rangle\,|a \oplus b\rangle$, is the special case $U = X$. The description "apply U if the control is 1" is exact only for control states $|0\rangle$ and $|1\rangle$. When the control is in a superposition, linearity produces a superposition of the two branches, and the output is in general entangled; the Bell preparation circuit is precisely this phenomenon with $U = X$. **??** shows that for suitable targets the roles are reversed and the control qubit acquires a phase, an effect called phase kickback on which the algorithms of the course depend.

The controlled-Z gate $C(Z) = \mathrm{diag}(1, 1, 1, -1)$ is symmetric between its two qubits, so for it the words control and target are merely labels.

In block form $C(U) = \mathrm{diag}(\mathbb{I}, U)$, so $C(Z) = \mathrm{diag}(1, 1, 1, -1)$, which is unchanged if the roles of control and target are exchanged. Conjugating the target by a unitary V gives $(\mathbb{I}\otimes V)\, C(U)\, (\mathbb{I}\otimes V^\dagger) = C(VUV^\dagger)$, since $|0\rangle\langle 0| \otimes V\mathbb{I}V^\dagger + |1\rangle\langle 1| \otimes VUV^\dagger$. With $V = H$ and $U = X$, the identity $HXH = Z$ from **??** gives $C(Z) = (\mathbb{I} \otimes H)\,\mathrm{CNOT}\,(\mathbb{I} \otimes H)$: a controlled-$Z$ is a CNOT with Hadamards on the target.

1.1.3 Other gates

The following gates appear in later chapters and in the homework. They are defined by their action; the matrices are recorded below, and nothing here is derived.

Gate	Action on basis states	Notes
Y	$\|0\rangle \mapsto \mathrm{i}\,\|1\rangle$, $\|1\rangle \mapsto -\mathrm{i}\,\|0\rangle$	third Pauli gate, $Y = \mathrm{i}XZ$
S	$\|0\rangle \mapsto \|0\rangle$, $\|1\rangle \mapsto \mathrm{i}\,\|1\rangle$	phase gate, $S^2 = Z$
T	$\|0\rangle \mapsto \|0\rangle$, $\|1\rangle \mapsto \mathrm{e}^{\mathrm{i}\pi/4}\,\|1\rangle$	$T^2 = S$
SWAP	$\|a\rangle\,\|b\rangle \mapsto \|b\rangle\,\|a\rangle$	exchanges two qubits, equals three CNOTs
Toffoli	$\|a\rangle\,\|b\rangle\,\|c\rangle \mapsto \|a\rangle\,\|b\rangle\,\|c \oplus ab\rangle$	doubly controlled X, a reversible AND

Matrices. The one-qubit gates of the table are

$$Y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e}^{\mathrm{i}\pi/4} \end{pmatrix}, \tag{1.4}$$

and, in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\mathrm{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.5}$$

Direct multiplication gives $S^2 = Z$, $T^2 = S$, and $\mathrm{i}XZ = Y$. The gate SWAP is a product of three CNOT gates with alternating roles, $\mathrm{SWAP} = \mathrm{CNOT}_{12}\,\mathrm{CNOT}_{21}\,\mathrm{CNOT}_{12}$, where CNOT_{21} has qubit 2 as control and qubit 1 as target. On basis states,

$$|a\rangle\,|b\rangle \xmapsto{\mathrm{CNOT}_{12}} |a\rangle\,|a \oplus b\rangle \xmapsto{\mathrm{CNOT}_{21}} |b\rangle\,|a \oplus b\rangle \xmapsto{\mathrm{CNOT}_{12}} |b\rangle\,|a\rangle\,. \tag{1.6}$$

The Toffoli gate is the 8×8 permutation matrix that exchanges $|110\rangle$ and $|111\rangle$ and fixes the other six basis states; it is $C(C(X))$, a controlled-CNOT.

1.1.4 Universality

A natural question is whether a fixed finite collection of gates suffices to build every possible quantum computation. Exact synthesis from a finite set is impossible, since such a set generates only countably many circuits whereas there are uncountably many unitaries, so the right notion is approximation.

Definition 1.2: Approximate universality

A finite set G of gates is universal if for every number of qubits n, every unitary V on n qubits, and every tolerance $\varepsilon > 0$, there is a circuit built from gates in G, each applied to some of the n qubits, whose overall unitary W satisfies $\|V - \mathrm{e}^{\mathrm{i}\gamma}W\| < \varepsilon$ for some global phase $\mathrm{e}^{\mathrm{i}\gamma}$. Here $\|\cdot\|$ is the operator norm, defined below.

The norm in Definition 1.2 is the operator norm, $\|A\| = \max_{\|v\|=1}\|Av\|$, the largest factor by which a matrix can stretch a unit vector; two unitaries that are close in this norm produce nearly the same measurement statistics on every input.

The set $\{H, T, \text{CNOT}\}$ is universal in this sense. We state this without proof. Three remarks keep the statement of universality honest. First, the gate T cannot be dropped: circuits built from H, S, and CNOT alone generate, up to global phase, only a finite set of unitaries on any fixed number of qubits, so they approximate nothing they do not reach exactly. Second, universality is a statement about possibility, not about cost. A generic unitary on n qubits requires a number of gates exponential in n, and the question of which unitaries admit short circuits is the subject of quantum algorithms, beginning in **??**. Third, for a single qubit the number of gates from $\{H, T\}$ needed to reach tolerance ε grows only as a power of $\log(1/\varepsilon)$, which is why approximation is practical. Synthesis results such as the Solovay-Kitaev theorem, which establish this bound, are outside this course [1].

1.2 No-cloning

Classical information can be copied without limit, and every classical computer does so constantly. It is natural to ask for a circuit that copies a qubit: a fixed unitary that takes an unknown state $|\psi\rangle$ and a blank qubit in some standard state $|s\rangle$ and returns two qubits each in the state $|\psi\rangle$. No such circuit exists, and the reason is nothing more than linearity [2, 3].

Theorem 1.1 No-cloning

There is no unitary U on two qubits and no fixed state $|s\rangle$ such that $U(|\psi\rangle \otimes |s\rangle) = |\psi\rangle \otimes |\psi\rangle$ for every qubit state $|\psi\rangle$.

Proof. Suppose such a U and $|s\rangle$ existed. Applying the cloning property to the two computational basis states gives $U(|0\rangle|s\rangle) = |0\rangle|0\rangle$ and $U(|1\rangle|s\rangle) = |1\rangle|1\rangle$. Now apply U to $|+\rangle|s\rangle$. Since U is linear,

$$U(|+\rangle|s\rangle) = \frac{1}{\sqrt{2}}\left(U(|0\rangle|s\rangle) + U(|1\rangle|s\rangle)\right) = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) = \left|\Phi^+\right\rangle. \tag{1.7}$$

But the cloning property applied to $|+\rangle$ itself demands

$$U(|+\rangle|s\rangle) = |+\rangle \otimes |+\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right). \tag{1.8}$$

The two vectors differ, for instance in the amplitude of $|01\rangle$, so no such U exists. ∎

The proof uses nothing about U beyond linearity, and the same argument extends to any dimension and to a copying device with auxiliary qubits, provided one allows the auxiliary system to end in a state that depends on the input and draws the contradiction from a component, such as $|01\rangle$, that the two sides must give with different weights. What it shows is that a single device cannot copy every state. It says nothing against copying states that are known, or states drawn from a known orthogonal set.

Wrong Concept 1.1: No-cloning forbids copying $|0\rangle$ and $|1\rangle$

The gate CNOT with a blank target does copy the computational basis states: $|a\rangle|0\rangle \mapsto |a\rangle|a\rangle$ for $a \in \{0, 1\}$. This is exactly classical copying, and it is allowed. The theorem forbids one fixed circuit that copies every input, and the proof shows where the attempt fails: fed $|+\rangle$, the same CNOT produces the entangled state $|\Phi^+\rangle$ rather than the product $|+\rangle|+\rangle$. Similarly,

anyone who knows a state can prepare as many copies of it as they wish. The restriction concerns states that are unknown to the person holding them.

No-cloning has direct consequences. An unknown qubit cannot be measured by first making many copies and measuring each in a different basis, so the amplitudes α and β of a single unknown qubit are not accessible; one measurement yields one bit, and the state is then gone. It also shapes the question of communication. If a state cannot be duplicated, can it at least be moved from one place to another without physically carrying the qubit? Teleportation answers yes, provided the two parties share entanglement and can exchange classical bits [4].

1.3 Teleportation

The setting is as follows. Alice holds a qubit, which we call qubit 1, in a state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ that neither she nor Bob knows. Alice and Bob also share a Bell pair $|\Phi^+\rangle$ on qubits 2 and 3, with qubit 2 at Alice's location and qubit 3 at Bob's. The pair was prepared earlier, by the circuit of **??**, and distributed before $|\psi\rangle$ existed. Finally, Alice can send classical bits to Bob. The goal is that qubit 3 ends in the state $|\psi\rangle$.

Definition 1.3: Quantum teleportation protocol

With qubits labelled as above, the protocol consists of four steps.

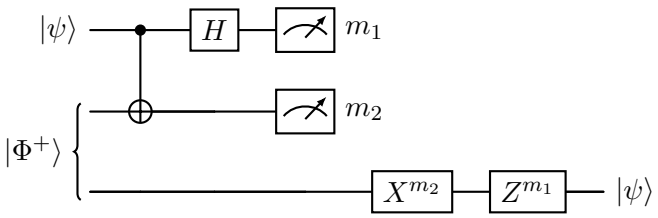
1. Alice applies CNOT with qubit 1 as control and qubit 2 as target, then applies H to qubit 1.
2. Alice measures qubits 1 and 2 in the computational basis, obtaining bits m_1 and m_2.
3. Alice sends the two bits $m_1 m_2$ to Bob over a classical channel.
4. Bob applies X^{m_2} and then Z^{m_1} to qubit 3, that is, he applies X if $m_2 = 1$ and Z if $m_1 = 1$.

At the end, qubit 3 is in the state $|\psi\rangle$, and qubits 1 and 2 are in the state $|m_1 m_2\rangle$.

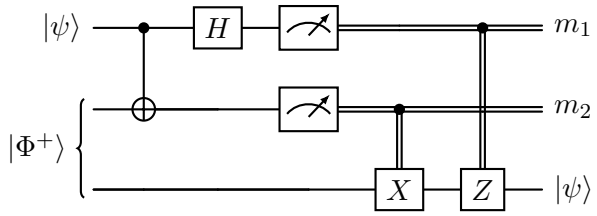
The resources deserve to be named precisely, because each plays a distinct role. One shared Bell pair supplies the correlation between the two locations; it is consumed, since at the end qubits 2 and 3 are no longer entangled with each other. Local operations at each location, two gates and a measurement for Alice, at most two gates for Bob, do the processing. Two classical bits carry the only message that crosses between the parties during the protocol. Nothing quantum travels from Alice to Bob after $|\psi\rangle$ is created, and neither party ever learns α or β. The word teleportation refers to the transfer of the state $|\psi\rangle$ from qubit 1 to qubit 3; the qubits themselves do not move, and no second copy of $|\psi\rangle$ is created anywhere, in agreement with Theorem 1.1.

1.3.1 The circuit

The protocol is the circuit below. Qubit 1 is the top wire, qubit 3 the bottom wire, and the gates X^{m_2} and Z^{m_1} on Bob's wire are controlled by the classical outcomes; the version with explicit classical wires is drawn after it.



Double lines denote classical bits. The gates X and Z on Bob's wire are applied conditionally on the classical values m_2 and m_1 respectively, so the circuit is the protocol of Definition 1.3 drawn in full.



Reading the diagram from left to right reproduces the four steps of the protocol: Alice's two gates, her two measurements, the two classical bits travelling to Bob, and Bob's two conditional gates. The vertical double lines are the only connections between Alice's wires and Bob's after the initial Bell pair, and they carry classical information.

1.3.2 The teleportation identity

The mechanism rests on a single algebraic identity. The initial three-qubit state is $|\psi\rangle_1 \otimes |\Phi^+\rangle_{23}$, and it can be rewritten by expanding qubits 1 and 2 in the Bell basis instead of the computational basis:

$$|\psi\rangle \otimes \left|\Phi^+\right\rangle = \frac{1}{2}\Big[\left|\Phi^+\right\rangle_{12} \otimes |\psi\rangle + \left|\Phi^-\right\rangle_{12} \otimes Z\,|\psi\rangle + \left|\Psi^+\right\rangle_{12} \otimes X\,|\psi\rangle + \left|\Psi^-\right\rangle_{12} \otimes XZ\,|\psi\rangle\Big]. \tag{1.9}$$

This is a statement about the state before anything has happened. Nothing has been measured and no gate has been applied; the right-hand side is the same vector as the left, regrouped. Read in this form, the state is a superposition of four terms, and in each term Alice's two qubits are in one of the four Bell states while Bob's qubit is in $|\psi\rangle$ acted on by one of the four operators $\mathbb{I}$, Z, X, XZ. The full expansion is written out in section 1.3.4, where the identity is verified term by term.

1.3.3 Bell-basis measurement

Alice's task is therefore to measure qubits 1 and 2 in the Bell basis. The circuit does this with the gates already available: CNOT followed by H on the first qubit maps $\left|\Phi^+\right\rangle, \left|\Psi^+\right\rangle, \left|\Phi^-\right\rangle, \left|\Psi^-\right\rangle$ to $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ respectively, so a computational-basis measurement after these two gates is a Bell-basis measurement of the original qubits. This is the inverse of the Bell preparation circuit, as the following calculation shows. Each Bell state has amplitude $\frac{1}{2}$ in (1.9), so each of the four outcomes occurs with probability $\frac{1}{4}$, independently of α and β.

Applying CNOT and then H on the first qubit to each Bell state:

$$\left|\Phi^+\right\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \xmapsto{\text{CNOT}} \tfrac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = |+\rangle\,|0\rangle \xmapsto{H\otimes\mathbb{I}} |00\rangle\,, \tag{1.10}$$

$$\left|\Psi^+\right\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \xmapsto{\text{CNOT}} \tfrac{1}{\sqrt{2}}(|01\rangle + |11\rangle) = |+\rangle\,|1\rangle \xmapsto{H\otimes\mathbb{I}} |01\rangle\,, \tag{1.11}$$

$$\left|\Phi^-\right\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \xmapsto{\text{CNOT}} \tfrac{1}{\sqrt{2}}(|00\rangle - |10\rangle) = |-\rangle\,|0\rangle \xmapsto{H\otimes\mathbb{I}} |10\rangle\,, \tag{1.12}$$

$$\left|\Psi^-\right\rangle = \tfrac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \xmapsto{\text{CNOT}} \tfrac{1}{\sqrt{2}}(|01\rangle - |11\rangle) = |-\rangle\,|1\rangle \xmapsto{H\otimes\mathbb{I}} |11\rangle\,. \tag{1.13}$$

Since the four Bell states are orthonormal and are mapped to the four computational basis states, a computational-basis measurement after these gates has outcome $m_1 m_2$ with probability equal to the squared modulus of the amplitude of the corresponding Bell state in the input, and this is what measuring in the Bell basis means. Conceptually, $(H \otimes \mathbb{I})\,\text{CNOT}$ is the inverse of the Bell preparation unitary $\text{CNOT}\,(H \otimes \mathbb{I})$ of section 1.1.1, since both factors are their own inverses; the measurement circuit undoes the preparation circuit.

1.3.4 Outcomes and corrections

Once Alice observes the outcome $m_1 m_2$, with m_1 the bit from qubit 1 and m_2 the bit from qubit 2, the projection rule leaves Bob's qubit in the corresponding term of (1.9). The correction relation is the following table; in every row the correction is $Z^{m_1} X^{m_2}$, which is step 4 of the protocol.

Outcome $m_1 m_2$	Alice's Bell state	Bob's state	Bob's correction
00	$\|\Phi^+\rangle$	$\|\psi\rangle$	$\mathbb{I}$
01	$\|\Psi^+\rangle$	$X\|\psi\rangle$	X
10	$\|\Phi^-\rangle$	$Z\|\psi\rangle$	Z
11	$\|\Psi^-\rangle$	$XZ\|\psi\rangle$	ZX

One branch is worked in the example below, and all four are traced in the full expansion that follows it. In the last row ZX and XZ differ only by a global phase, since $ZX = -XZ$, so either order of Bob's two gates recovers $|\psi\rangle$ up to an unobservable phase.

Example 1.1 (One measurement branch)

Suppose Alice obtains $m_1 m_2 = 01$. The outcome 01 corresponds to the Bell state $|\Psi^+\rangle$ on Alice's qubits, so by (1.9) Bob's qubit is in the state $X|\psi\rangle = \beta|0\rangle + \alpha|1\rangle$: the amplitudes of $|\psi\rangle$ with the roles of $|0\rangle$ and $|1\rangle$ exchanged. Since $m_1 = 0$ and $m_2 = 1$, Bob applies X and no Z, and $X(\beta|0\rangle + \alpha|1\rangle) = \alpha|0\rangle + \beta|1\rangle = |\psi\rangle$. The other three branches are traced below.

The point of the example is that the correction is determined by the two bits alone, without knowledge of α or β, and that in this branch, until Bob applies it, his qubit is in general not in the state $|\psi\rangle$.

Full expansion. Write the initial state with qubits 1 and 2 first and qubit 3 last:

$$|\psi\rangle \otimes |\Phi^+\rangle = \frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle). \tag{1.14}$$

The CNOT with control 1 and target 2 flips the second bit of the last two terms, giving

$$\frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle). \tag{1.15}$$

The Hadamard on qubit 1 then replaces $\alpha|0\rangle$ by $\alpha|+\rangle$ and $\beta|1\rangle$ by $\beta|-\rangle$ on the first qubit:

$$\frac{1}{2}\Big[\alpha(|0\rangle + |1\rangle)|00\rangle + \alpha(|0\rangle + |1\rangle)|11\rangle + \beta(|0\rangle - |1\rangle)|10\rangle + \beta(|0\rangle - |1\rangle)|01\rangle\Big]. \tag{1.16}$$

Collecting the eight terms according to the values of the first two qubits gives

$$\frac{1}{2}\Big[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\beta|0\rangle + \alpha|1\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(-\beta|0\rangle + \alpha|1\rangle)\Big]. \tag{1.17}$$

Each of the four outcomes $m_1 m_2$ has probability $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$, and the conditional states of qubit 3 are $|\psi\rangle$, $X|\psi\rangle$, $Z|\psi\rangle$, and $XZ|\psi\rangle = X(\alpha|0\rangle - \beta|1\rangle) = -\beta|0\rangle + \alpha|1\rangle$. The corrections $\mathbb{I}$, X, Z, ZX return each to $|\psi\rangle$; in the last case $ZX\,XZ|\psi\rangle = Z\,\mathbb{I}\,Z|\psi\rangle = |\psi\rangle$ exactly, while $XZ\,XZ|\psi\rangle = -|\psi\rangle$ differs by a global phase. Applying $(H \otimes \mathbb{I})\,\mathrm{CNOT}$ to the Bell states on qubits 1 and 2 in (1.9), using the mapping of section 1.3.3, reproduces (1.17) term by term, which verifies (1.9). Alternatively, substitute $|00\rangle = \frac{1}{\sqrt{2}}(|\Phi^+\rangle + |\Phi^-\rangle)$, $|11\rangle = \frac{1}{\sqrt{2}}(|\Phi^+\rangle - |\Phi^-\rangle)$, $|01\rangle = \frac{1}{\sqrt{2}}(|\Psi^+\rangle + |\Psi^-\rangle)$, $|10\rangle = \frac{1}{\sqrt{2}}(|\Psi^+\rangle - |\Psi^-\rangle)$ for the first two qubits in the initial state and collect terms.

1.3.5 No-signalling

Why is the classical message necessary? Before the bits arrive, Bob knows only that his qubit is in one of the four states $|\psi\rangle$, $X|\psi\rangle$, $Z|\psi\rangle$, $XZ|\psi\rangle$, each with probability $\frac{1}{4}$. Averaged over these four possibilities, every measurement Bob can perform gives each of its two outcomes with probability exactly $\frac{1}{2}$, whatever α and β are; this is verified below. It could not be otherwise: the no-signalling argument of **??**, applied to Alice's two qubits and to the basis that CNOT followed by H selects, says that nothing Alice does locally can change the statistics of any measurement Bob performs, and before the protocol Bob's qubit was half of a Bell pair, which gives uniformly random outcomes in every basis. The information about $|\psi\rangle$ is therefore split between two parts, Bob's qubit and Alice's two bits, and neither part alone reveals anything about α or β. The two bits themselves are uniformly random and independent of $|\psi\rangle$. Only their combination, the correction applied to the qubit, reconstitutes the state.

Bob's statistics. Let $\{|u\rangle, |v\rangle\}$ be any orthonormal basis for Bob's qubit. Averaged over the four equally likely branches, the probability of the outcome u is

$$\frac{1}{4}\sum_{P\in\{\mathbb{I},X,Z,XZ\}} |\langle u|P|\psi\rangle|^2 = \frac{1}{4}\,\langle u|\left(\sum_P P\,|\psi\rangle\langle\psi|\,P^\dagger\right)|u\rangle\,. \tag{1.18}$$

Write the outer product as $M = |\psi\rangle\langle\psi| = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a + d = |\alpha|^2 + |\beta|^2 = 1$. Then

$$XMX = \begin{pmatrix} d & c \\ b & a \end{pmatrix}, \qquad ZMZ = \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}, \qquad XZ\,M\,ZX = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}, \tag{1.19}$$

so the four matrices M, XMX, ZMZ, $XZ\,M\,ZX$ sum to $2(a+d)\,\mathbb{I} = 2\,\mathbb{I}$. The averaged probability is $\frac{1}{4}\cdot 2\,\langle u|u\rangle = \frac{1}{2}$ for every basis and every $|\psi\rangle$. For the computational basis this is visible directly from (1.17): the four branches give the outcome 0 with probabilities $|\alpha|^2$, $|\beta|^2$, $|\alpha|^2$, $|\beta|^2$, whose average is $\frac{1}{2}$. This is the calculation behind the claim that Bob learns nothing before the classical bits arrive, and it is the content of no-signalling in this particular protocol.

Wrong Concept 1.2: Teleportation sends matter or beats light

Nothing material travels from Alice to Bob. Qubit 3 was at Bob's location throughout; only its state changes. Nor is anything transmitted faster than light. Until the two classical bits arrive, Bob's qubit gives uniformly random results in every basis and carries no usable trace of $|\psi\rangle$, so the state becomes available to Bob exactly when the classical message does, and the classical channel is no faster than light. Teleportation also does not remove the need for a quantum channel: the Bell pair had to be distributed between the two locations beforehand. What it changes is when that channel is used, which may be long before the state to be sent exists, and how a fragile unknown state is handled, since only robust classical bits need to be transmitted once the pair is in place.

Remark 1.1 Resources consumed

After the protocol Alice's qubits are in $|m_1 m_2\rangle$, which carries no information about α or β, and the original copy of $|\psi\rangle$ is destroyed by the measurement, so exactly one copy exists at the end, as Theorem 1.1 requires. The Bell pair is spent, so a second state needs a second pair, and two classical bits are used; fewer cannot suffice, though we do not prove this. If qubit 1 is entangled with another system, that entanglement is transferred to Bob's qubit; this entanglement swapping is the basic operation of the quantum repeaters of **??**.

Remark 1.2 Information flow

The CNOT and H convert the question of which Bell state Alice's qubits are in into two classical bits; the Bell pair guarantees that Bob's qubit differs from $|\psi\rangle$ by one of four known operators; the classical channel tells Bob which one. Removing any ingredient breaks the protocol: without entanglement Bob's qubit is uncorrelated with $|\psi\rangle$, without the measurement no bits exist, and without the classical channel Bob cannot correct.

1.4 Summary

A circuit is a notation for composing gates: sequence is matrix multiplication in reverse order, and parallel placement is the tensor product. Controlled-U generalizes CNOT, and a superposed control in general produces entanglement rather than a conditional, the exception being a target that is an eigenvector of U. A finite gate set such as $\{H, T, \mathrm{CNOT}\}$ approximates every unitary to any tolerance, although exact synthesis from a finite set is impossible. An unknown quantum state cannot be copied, by linearity alone; known states can. Teleportation moves an unknown state from one qubit to another using one shared Bell pair, local gates and measurements, and two classical bits. Bob's qubit ends in a Pauli-rotated copy of the state, and the two bits identify which rotation to undo. The Bell pair is consumed, the original is destroyed, and nothing about the state reaches Bob before the classical message does, so no-signalling is respected.

Homework 2, part A, can be attempted after this lecture.

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